Palaeomagnetism of Plio-Pleistocene Basalts from North-west Sardinia

THERE is a growing body of evidence which supports the hypothesis of a counterclockwise rotation of Corsica and Sardinia with respect to southern France, in particular the palaeomagnetic data from the Permo-Carboniferous of Corsica[1,2] and the Permian and Oligo-Miocene of Sardinia[3-4]. Although it has been suggested that the rotation possibly took place as early as the Mesozoic[2], it is now more generally regarded as having occurred at some time during the Tertiary[6,7].

In 1971 we collected a preliminary suite of block samples from seven sites in the Plio-Pleistocene basalts of Logudoro in north-western Sardinia (Fig. 1) in an attempt to establish the date of completion of the rotation. The results from this suite of samples, as reported here, closely parallel results recently published by Bobier and Coulon[5] and by Manzoni *et al.*[8].

Much of Logudoro is underlain by Oligocene to Lower Miocene trachytic and andesitic volcanics with westerly palaeomagnetic declinations[4] (Table 2, collections 1, 2), and by Langhian to Tortonian marine and continental sedimentary rocks[9]. An erosion surface cut on the latter is overlain by a first generation of basalts, the *basalti degli altopiani* (β_1). These are usually only one flow thick, and cap conspicuous plateaus and mesas. Valleys eroded below the level of the basalt plateau are occupied by a younger generation of basalts[10] (β_2). At least some of the basalts of the second generation must be late Pleistocene or Recent, for they are associated with cinder cones almost untouched by erosion. These late basalts were preceded by a period during which erosion removed much of the *basalti degli altopiani*, so the latter are regarded as Pliocene or possibly Late Miocene in age.

The block samples were cored, and single cores from each block were subsequently measured and magnetically cleaned in the laboratory. We found that the minimum scatter of the remanence directions was obtained after cleaning in alternating fields of 300 oersted, and such directions were regarded as giving the best representation of the magnetic field of the site at the time of emplacement of the lavas. The results are given in Table 1.

In Table 2 we list chronologically the mean site data for all published Tertiary and Quaternary collections from Sardinia. In the computation of the site mean from the present study we calculated a mean for the normally magnetized β_2 basalts (using "normal" in the sense defined by Wilson *et al.*[11]). The very close agreement with the results of Bobier and Coulon[5] is satisfying in view of the dispersion of the directions within flows (as indicated by the high values of α_{95}). Because one of the three β_1 sites (site 3) has an intermediate direction[11] no β_1 mean was calculated.

Table 1 Parameters of the Cores

Site and unit	D	I	α_{95}	R	K	N
Site 1 (β_1)	16°	+66°	35.7°	3.604	7.6	4
Site 3 (β_1)	242°	−67°	6.1°	5.985	120.2	6
Site 5 (β_1)	344°	+62°	9.5°	5.902	51.1	6
β_1 mean:	Not calculated					
Site 2 (β_2)	2°	+55°	12.5°	2.979	98.5	3
Site 4 (β_2)	334°	+62°	27.9°	3.746	11.8	4
Site 6 (β_2)	341°	+31°	11.6°	5.854	34.3	6
Site 7 (β_2)	3°	+54°	12.4°	3.946	56.1	4
β_2 mean:	350°	+51°	18.6°	3.881	25.3	4

Inspection of Table 2 and Fig. 2 shows that the means of the post-Miocene basalts from different parts of Sardinia (collections 5–11) cluster around the present field direction[4] ($D=0°$, $I=60°$). This agrees with the previous conclusion that rotation had ceased before the end of the Late Miocene, which was based on the presence of Upper Miocene evaporites throughout the Ligurian sphenochasm, a basin that apparently opened behind the rotating block[6]. The view that rotation ceased before the end of the Late Miocene has received further support based on structural styles in Tuscany[12].

Declinations from the Oligocene–(?)Lower Miocene trachyandesites are displaced considerably to the west of the present declination. This led De Jong *et al.*[4] to conclude, with some reservations, that Sardinia had rotated after the middle Tertiary, and later work on the same rocks by Bobier and Coulon[5] gave similar palaeomagnetic results. We note, however, that the α_{95} values for collections 1 and 2 are high, so the angle of rotation is not well established.

The data from the β_1 basalts, as currently presented, cannot be used to substantiate or disprove the suggestion of Manzoni *et al.*[8] that the β_1 basalts might predate the end of rotation. We have therefore combined the β_1 site means from previous sources[5,8] with our data, rejecting sites with clearly spurious directions, and computed a single overall mean based on seven sites (B2, B3.1, B3.2 (ref. 5), Cantoniera and Lei[8], and our sites 1 and 5). In the same way an overall β_2 mean was computed

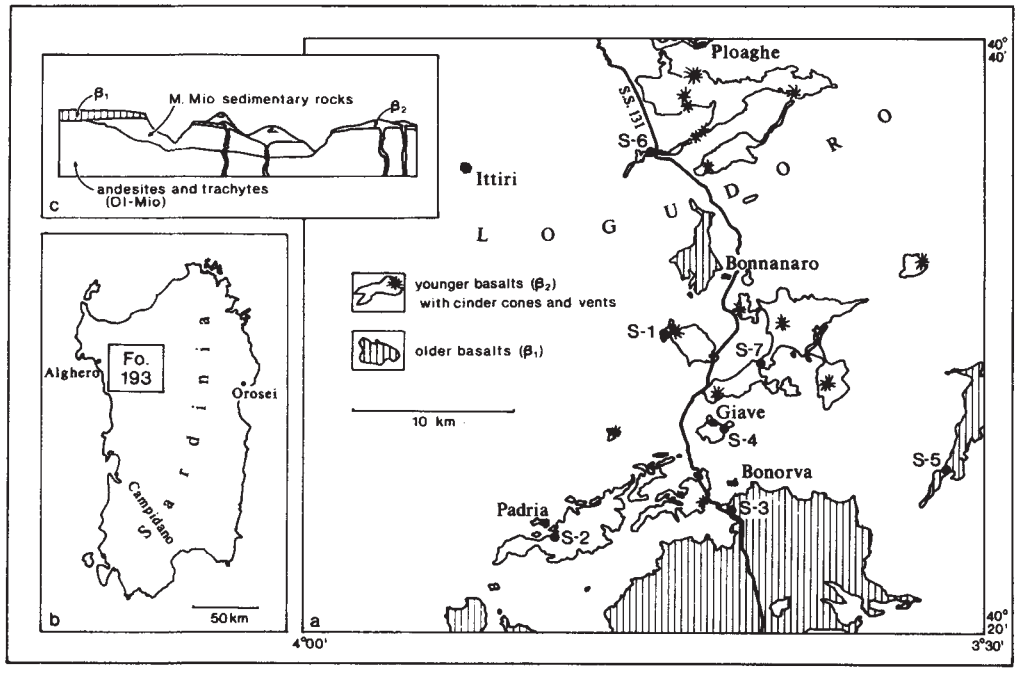


Fig. 1 *a*, Foglio (sheet) 193 "Bonorva" showing site locations and basalt outcrops according to Moretti *et al.*[9]. Longitudes are west of Rome, Monte Mario (12° 27′ 08.40″ E of Greenwich). *b*, Location of Foglio 193. *c*, Generalized stratigraphy (after Segre[9]).

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Table 2 Mean of Site Means

Collection	Age/locality	Lithology	D	I	α_{95}	R	K	No. of sites	Reference
11	Quaternary (Cedrino mean)	Basalt	181°	−62°	—	5.70	17	6	8
10	Quaternary (Cala Gonone mean)	Basalt	199°	−50°	—	3.99	280	4	8
9	Quaternary (Plasmasera mean)	Basalt	174°	−51°	—	3.00	547	3	8
8	Quaternary (Cala di Luna mean)	Basalt	180°	−55°	—	1.99	121	2	8
7	Pleistocene-Recent (?) (β_2 Logudoro mean)	Basalt	353°	+53°	16°	3.837	38	4	5
6	Pleistocene-Recent (?) (β_2 Logudoro mean)	Basalt	350°	+51°	18.6°	3.881	25.3	4	This communication
5	Post-Miocene (Campidano mean)	Basalt	183°	−51°	—	1.99	194	2	8
4	Post-Tortonian (β_1 Campeda mean)	Basalt	342°	+50°	—	1.99	147	2	8
3	Post-Tortonian (β_1 Logudoro mean)	Basalt	334° 30′	+66°	31°	3.320	185	4	5
2	Oligocene-(?) Lower Miocene (Alghero mean)	Trachyandesite	332°	+42°	11°	9.541	19.6	10	Calculated from ref. 4
1	Oligocene-(?) Lower Miocene (Logudoro mean)	Trachyandesite	335° 30′	+43° 30′	18°	9.607	13	11	5
	Overall mean for β_2		352.3	+48.4	9.3°	6.861	43	7	5 and this communication
	Overall mean for β_1		353.8	+56.0	8.09°	6.894	56.7	7	5, and 8 this communication

(B4, B6, B7 (ref. 5), and our sites 2, 4, 6, 7). Both overall means are given in Table 2, and Fig. 2. The mean directions calculated in this way are clearly inconsistent with rotation after cooling of the basalts. The absolute ages of the β_1 and β_2 units are not known with certainty because of problems with atmospheric argon contamination and the low potash content, of the lavas (M. Nicoletti and M. Fornaseri, personal communication).

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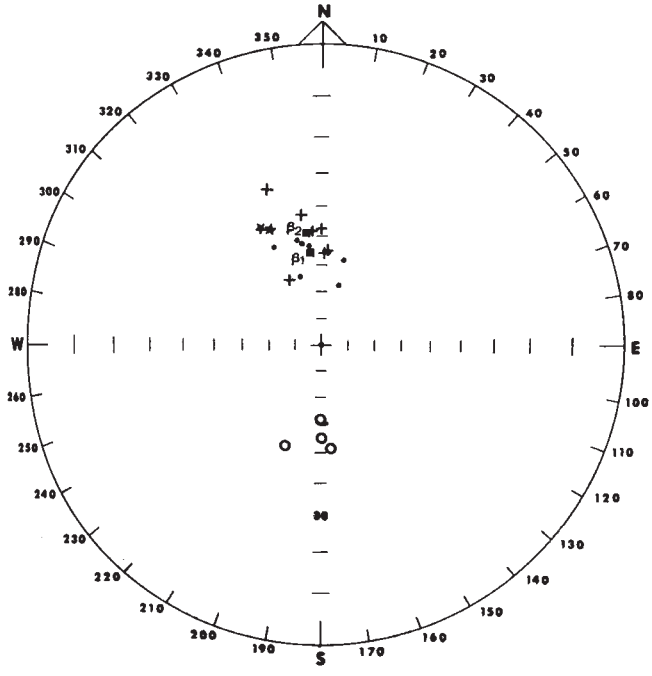


Fig. 2 Lower hemisphere plot of data from Table 2. ○, Quaternary basalt means (collections 8–11, all reversed); ■, β_1 and β_2 overall means; ●, β_1 data; +, β_2 data used in computing these means; *, Oligo-Miocene trachyandesite means (collections 1 and 2).

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Dislocations and Bubbles in Vein Quartz

Two types of quartz veins occur in the former goldfield area at Ballarat, Victoria; one is potentially gold bearing and the other had always been barren. The former (type B1 veins) which go deep into the crust are hydrothermal veins or quartz dykes associated with local granites. The latter (type B2

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